

Cybersecurity and Games

Learning through experiences and simulations

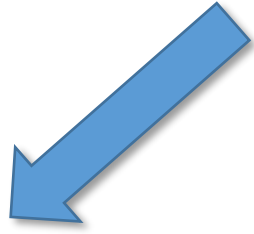
Plan

- I) Vulgarisation**
- II) How to build a game**
 - A) Test and critics**
 - B) Define the Objective**
 - C) Define the Gameplay**
 - D) Game elements**
 - E) Playtest and Iteration**
- III) Exemples**

But first, one Example



Find the LuxTrust mail



Spam

- Unsolicited advertisement
- 10 of them



Phishing

- Trying to steal information
- 7 of them



SpearPhishing

- Trying to steal information using personal informations
- 2 of them

Pros

Cons

What you have learned

Define the target audience

- If you want to communicate, you always need to adapt
 - Everyone has their experience, language specificities, references...
 - Define common characteristics of your audience.
 - Adapt the vocabulary and the examples...
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- Examples:
 - On your opinion, who is the target of the mail game ?
 - What would you do to adapt it to a younger audience ?

The mean of convey is also different

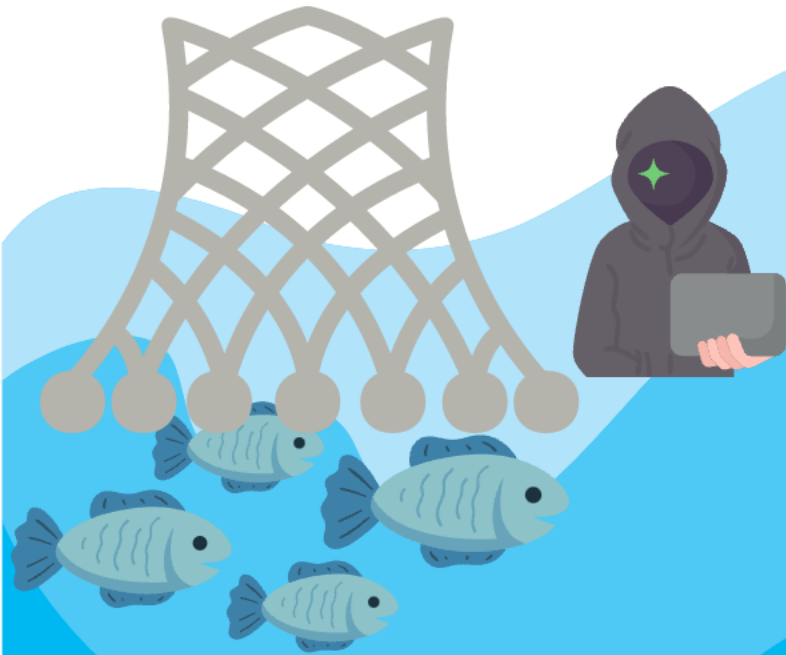
- Texts, Videos, Audio, Trainings, Images, Stories ...
- Each of them need to be thought differently
- Build around their habits, their daily consumption
- Examples: Symmetric vs Asymmetric cryptography
 - For kids: I would do an activity with a key and a chest to unlock.
 - For teenagers: I would build an exemple around TikTok or Roblox.
 - For adult: I would build an exemple around series (Breaking Bad, Star Wars...)

Vulgarisation

Language tips

- You should not use specialised vocabulary (exception to the learned notion) – **examples**: GDPR, BYOD, Pentest, UPS, Zero day ...
- Use analogy and metaphors, even when they are not exactly representative – **example**: Pipe leaking to represent a crisis
- Use Storytelling, The Hero's Journey of Joseph Campbell, also known as the monomyth.
- You can also use humor; be careful though, too much can undermine your lesson.

Phishing



 Not targeted !

 Global weaknesses



By Mail



By phone



with links



with attachments

Spearphishing



Targeted !



Investigation on
online presence



Observation

Vulgarisation

Visualisation

- You can use diagram , infographic or image to explain – **example**
Explain phishing or spearphishing with one picture.
- Chunking into smaller parts – **example**: “The rule of 6” in powerpoint slides
- Step-by-step reveal – **example**: Disclosing information with you to accompany it, like this presentation.
- Play with high contrast.

Engaging the audience

- Use some callbacks to create a sense of familiarity.
- Start with an ice breaker – **example**: The game played at first was an ice breaker; An activity that helps building something with the audience.
 - And this was a callback **example**.
- You can use ambiance (visual cue, sounds), and stories to give some emotion – **example**: talk about kids threat to young parents ...

Cognitive tricks

- Always start with something concrete, and then you can talk with theories – **examples**: In board games, it is more easy to start with a “fake turn” than heavy rules.
- Numbers should be translated to relatable examples – **example**: Password takes 3 trillion years -> If you count 1 number per second, it represent 95.000 years of counting.
- Use Storytelling, The Hero's Journey of Joseph Campbell, also knowed as the monomyth.
- You can also use humor; be careful though, too much can undermine your lesson.

Build a game

“In order to write about life first you must live it.” – Ernest Hemingway

Test & Critics

- Play all kind and sort of games ;
 - Board Games (Card, Dice, Tiles, Sports ...)
 - Video Games (Action, Adventure, RPG, Simulation, Puzzle ...)
 - Immersive Games (Escape Games, Tabletop RPG, LARP ...)
 - Gamification (Educational Games, Training Simulation, Gamified experience)
 - And all hybrid forms
- For each of them, ask yourselves some questions

Remember this ?

Pros

- What was fun ?
- What kept you going ?
- When did you felt excited ?

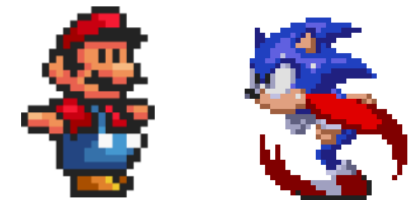
Cons

- What was boring ?
- When did you felt you wanted to stop ?
- What element was too long or boring ?

What you have learned

- What do you think you have understood ?
- Did the experience felt forced, or was it natural ?
- Was there something missing ?

- What makes a good game is the frustration of something missing – **example**: Yuji Naka created Sonic the Hedgehog because he thought Super Mario Bros. was too slow.



Build a game

Define the objective

- What you want players to learn ?
 - **Vocabulary** ? You will have to think about how to introduce it.
 - **Notions** ? You will have to think about how players can test them.
 - **History** ? You will have to think about your storytelling.
 - **Best Practices** ? You will have to think about how it can become a reflex. (*[My email game started here](#)*)
 - **Roles** ? You will have to think about how to make players adopt perspectives
 - **Ethics** ? You will have to think about how to bring dilemmas into play
- And of course, you can combine them !

Build a game

Define the objective

- Emotion and Experience
 - What are the feels you want your players to experiment ?
Examples: amuse, challenge, surprise, empower, reflective, fear...
 - The type of game might depend on that.
- Time and Context
 - How much time do you want the experience to last ?
 - The type of game might depend on that.
- Final message
 - Resume your goal with one sentence ; Use the KISS method.
Keep It Simple and Stupid.

Build a game

Define the Gameplay

- Understand the notion of Gameplay loop
 - This is the serie of actions your repeat to play the game

Example:

- Play a card if possible
 - If not, draw a card
 - Play the card if possible
- Define all the differents roles (competitve, cooperative ...)
 - Example:** Werewolf (Villager, Werewolf, Witch ...)
- Define the win or loose conditions
 - To keep the game engaging, it shouldn't be easy to guess what the end will look like.



Build a game

Define the Gameplay

- Use existant gameplay and give a coat of Cybersecurity
 - It is easy to modify
 - It is difficult to have a learning experience

example: Malwopoly

- Inspire yourself from existant gameplay and modify it
 - It is easy for the player to learn
 - It is difficult to break the balance

example: Balatro

- Create a new gameplay from scratch
 - It is easy to put learning as a main goal
 - It is difficult to have a fun game



Build a game

Game Elements

- There are two types that you will need to define
 - The elements that you will provide

Examples: Cards, Dices (exotic, such as D20, D100), Game Board, Assets, Graphics, Texts ...
 - The elements that you need to rely on the user to have

Examples: Dice (non exotic, such as D6), Coin, Tablet/Phone ...
- The element you provide can be customisable and should always be there.
- The element that you do not provide should be considered as comfort, not mandatory.
 - If they are mandatory, it should be clearly stated.

Build a game

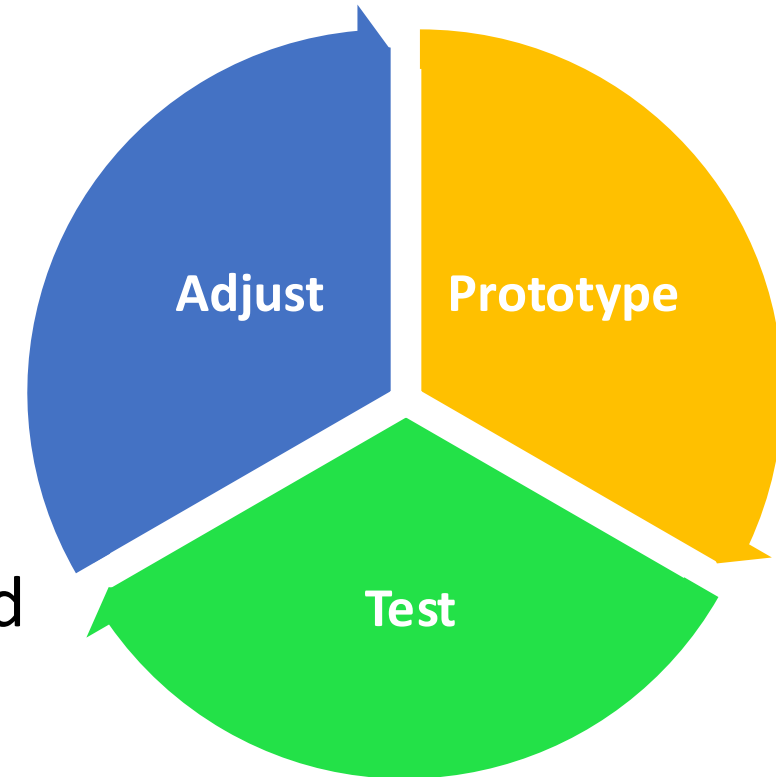
Game Elements

- Is there a time constraint in your gameplay loop ?
 - You can use phone, an programmed app
 - You can also provide a sandglass
- Is there a need to represent someone/something ?
 - You can use a token, a pawn
- Is there some chance elements ?
 - You can use a coin, a dice, a spinner ...
- Do you have some ressources counts ?
 - You can use tokens, markers, sheets
- Is there a need to reprsent a place, or place precisely elements ?
 - You can use a map or a board

Build a game

Playtest and Iterations

- The best way to test a game : play it!
- And to play it, you need to build **prototypes**
 - Use some easy material
 - Papers, cardboard, post-it ...
 - Goal: make it work, understandable
- **Test** yourself, make the other test
- **Adjust** your prototype with what you have learned



Example

- A normal lost phone
- CyberDuel
- Cyber Wargame
- Malwopoly
- Hackenators
- The Password Game
- Malware & Monsters
- Code Hunters
- Cybersecurity Fresque
- CyberStrategia
- root-me.org
- Hackathon
- Red Team vs Blue Team
- ...

Format	Examples (Educational Cybersecurity Games)
Board/Card Games	CTD, Cyber Attack Chain, Crypto Go, Backdoors & Breaches, Cyber Threat Guardian, etc.
Digital / Online Games	KC7, OverTheWire, TryHackMe, Targeted Attack, Cyber Awareness Challenge, etc.
Simulation / VR	CyberCIEGE, HackOps, Deepspace Danger
Serious / Academic Games	SherLOCKED, PeriHack, CyberMoraba, Phishing-writing simulations

Questions ?

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